PROBLEM STATEMENT 01 / SOLUTION DECK

PRISM

Property Research, Intelligence & Smart Modeling

An auditable property valuation engine for India

01 EVIDENCE

Unify fragmented property data

02 ESTIMATE

Price the asset and show uncertainty

03 REVIEW

Route the case with an audit trail

Pune apartment pilot / Team PRISM

Start with Pune apartments

Lenders must reconcile conflicting records, inconsistent area definitions and uneven local evidence.

FIRST MARKET

Baner + Kharadi

Pilot for residential apartments. Baner has a live ingestion test. Kharadi is enabled for discovery and is the next pilot check.

RESIDENTIAL EXPANSION

State by state

Expand across Pune, then test other metros. Add registry adapters and validate local coverage, error and interval calibration before launch.

COMMERCIAL DESIGN

Separate model

Start with office and retail. Use lease comps, vacancy and cap rates. Cross-check stabilized net operating income ÷ cap rate against sale comps.

Success means faster lender review with measurable valuation error and clear evidence gaps.

The MCP makes evidence reusable

A property request becomes a source-linked evidence package for valuation and lender review.

INPUT

Property request

Address, property type, area basis and case date.

→

WORKING PROTOTYPE

Evidence MCP

Discover sources, normalize records, flag conflicts and select comparable candidates.

→

PROPOSED

Laya + valuation

Score evidence, estimate value from verified comps and calibrate a range.

→

PROPOSED

Review report

Return factors, source IDs, risk flags and a routing recommendation.

Go MCP / Rust reconciliation / Python bridge + geodata / Next.js / React viewer

Keep each source in its proper role

Keep source URLs, retrieval times, event dates and immutable snapshots with every record.

SOURCE	USE AND TRANSFORMATION	PILOT STATUS
IGR Maharashtra + lender deeds	Normalize registered sale price and area. Use government reference rates as separate features.	**Access pending**
MahaRERA + property inputs	Join project ID and attributes. Flag disagreement on area or address.	**Project lookup**
Public listings via Firecrawl	Extract asking price, area basis and date. Track freshness and local asking-price trends.	**Live Baner audit**
Overture + Sentinel-2	Compute nearby amenities and distances. Add vegetation and built-up proxies from cloud-masked imagery.	**Map live / imagery planned**
Maha Metro + public notices	Geocode dated projects. Keep announced and operational infrastructure separate.	**Planned**

Match by locality and project, then check unit identifiers, area basis and dates. Keep unresolved records separate.

What the Baner run proved

The MCP discovered sources and ingested individual listing pages through local Firecrawl and Go parsing.

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MCP TOOLS

Discovery, ingestion, geocoding, evidence lookup, reconciliation and comp selection.

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REAL LISTINGS INGESTED

One portal. Each listing retained an asking price and a retrievable source snapshot.

0

ELIGIBLE EXAMPLE COMPS

The parser did not capture area or publication date. The comp selector returned no matches.

11 Nov 2026 IST / Baner, Pune / Example subject uses 1,000 sq ft of carpet area

Prove the lift on held-out sales

Acquire and audit completed-sale records before training or claiming valuation accuracy.

BENCHMARK

Start with comps

Use median registered sale price per sq ft with consistent area basis. Add a hedonic benchmark and the lender’s incumbent AVM if available.

CANDIDATE

Train CatBoost

Use sale comps, property attributes, geodata and market signals. Test the lift from Laya evidence scoring and deduplication.

VALIDATION

Split by date

Train on older sales, calibrate on a later window and test on the newest. Group repeated units and use only evidence known by the cutoff.

EVALUATION SCORECARD

MAE in INR / MAPE / % within ±10% / empirical coverage of 90% conformal ranges

Compare on identical cases. Report sample size, interval width, abstention and error by locality.

Show the lender the evidence

A versioned report separates the estimate, uncertainty, contributing factors and source quality.

SAVED EVIDENCE EXAMPLE

Baner apartment, Pune

₹2.90 crore

Publisher asking price

Area and publication date missing in extracted record.

Case status / Insufficient comparable evidence

Source snapshot pkg_ad1907b89d90…

PROPOSED VALUATION REPORT

Structured output

Value and calibrated range, when supported.

Top SHAP factors, comp IDs, evidence quality and model version.

A narrative reads the JSON. Validate every number and source reference before display.

Route the case by evidence and risk

Policy checks run first. PRISM recommends a review route and records the reason.

01	**Strong evidence + narrow range**	Desktop review if lender policy permits
02	**Mixed evidence or wider range**	Analyst review and request missing evidence
03	**Material conflict, weak evidence or mandatory check**	Physical valuation through the lender’s process

Pilot measures

Track physical referrals, missed high-risk cases, error by route and median and p95 turnaround in shadow mode.

Stress-test LTV using the lower end of the range. Lender policy governs approved collateral values, mandatory inspections and valuer sign-off. See RBI HFC Directions, para. 215.

Scale only after the pilot passes

Reuse the evidence schema and MCP tools while validating each new source and market.

1 / EVIDENCE PILOT

Make the data usable

Audit dates and area bases. Add authorized sale labels. Record source rights and measure completeness in both pilot localities.

2 / SHADOW PILOT

Measure the workflow

Run alongside lender decisions. Cache geodata, queue source jobs, measure latency and monitor missing fields, drift and interval coverage.

3 / EXPANSION

Add markets in stages

Add adapters for registries and city feeds. Calibrate by locality and asset class. Launch commercial only after lease and sale data pass review.

Release gates / Authorized labels / Holdout lift / Interval coverage / Lender sign-off

Sources and verification

External references are clickable. Prototype results come from the team’s saved run artifacts.

Challenge brief and organizer listing
Problem Statement 1 and six requirements

Magicbricks Baner listing
Source for the example asking-price record

MahaRERA
Project registry and disclosures

IGR Maharashtra
Candidate source for registered records and reference rates

Overture Places
Mapped amenities and source attribution

Copernicus Sentinel-2
Proposed satellite features

Maha Metro Pune project profile
Dated transport and project context

RBI HFC Directions, 2025
LTV and valuation governance reference

Team artifacts / Baner live audit, 11 Nov 2026 IST / MCP demo verification, 7 Nov 2026

See the supplied challenge brief, pp. 1–3, for the complete judging criteria and deliverables.